TECHNICAL ADDENDUM

Hawking Evaporation Flux & Nondissipative Reactive Energy Exchange at the Retrograde Front Boundary

A conjectural boundary-condition extension for the QVEX NGC 6503 framework

Document ID	QVEX-ANNEX-A-HF-01 (QVEX-TH-RETRO-01)
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Date	October 2026
Epistemic status	Analytical Hypothesis / Boundary Condition Extension; conjectural and unverified against SPARC residuals per Annex A2 §1.

Abstract & executive summary

This addendum proposes a boundary-level description of a transition from dissipative-friction language to reactive, non-dissipative energy storage in a hypothesized quantum-vacuum foam around NGC 6503. The model premise is a balance between leading-front compressive pressure (فشردگی جبهه پیشرو خلا) and trailing-front vacuum-density-deficit suction (مکش از خلا حاصل از کسری تراکم جبهه پشتی), with a central black-hole Hawking flux serving as a proposed driver. These are model assumptions, not established mechanisms or observational findings.

In the proposed picture, energy transferred at the leading front is retained as effective momentum/persistent kinetic energy, while a trailing deficit is represented through a foam-induced metric perturbation. The radial laws below specify a phenomenological profile and expose its sign-changing raw friction proxy. They do not constitute a derivation from semiclassical gravity, nor a fit to rotation-curve residuals.

Scope & epistemic boundary. This document records an analytical hypothesis only. The quoted parameter values, Hawking coupling, front forces, and PID mapping are not independently validated. In particular, SPARC residual validation is outstanding (Annex A2 §1). Interpret all expressions as testable ansätze, not measurements.

1. Physical mechanism: reactive virtual friction

“Virtual friction” is used here as a descriptive label for an exchange process, not as ordinary thermodynamic drag. The defining idealization is zero dissipative heating; energy transferred into the moving component is stored reversibly in its effective momentum/kinetic state. Any real implementation would need a covariant action or constitutive law specifying the degrees of freedom, exchange current, and stability conditions.

$$\text{Non-dissipative idealization:}\quad P_{diss} = 0$$

$$\text{Reactive momentum bookkeeping:}\quad \delta m_{eff}\, u^{\mu}$$

The trailing-front deficit is represented phenomenologically by an effective spacetime perturbation $h^{foam}{}_{\mu\nu}$. This notation does not supply field equations or establish that the perturbation is generated by a Hawking flux; those are open derivation requirements.

Covariant bookkeeping and closed-orbit condition

$$\nabla_{\mu} \left(T^{\mu\nu}{}_{matter} + T^{\mu\nu}{}_{reactive_foam} \right) = 0$$

$$\oint F_{retro} \cdot dr = 0$$

The conservation relation is a required total-system constraint. If matter and foam exchange energy-momentum separately, the corresponding sector tensors need not be independently conserved; equal and opposite source currents must then be specified. The closed-orbit integral states a conservative-cycle idealization and is not implied merely by setting P_{diss} to zero.

Interpretation of “fronts”

The leading and trailing labels identify opposite sides of a moving disturbance. The model assigns positive compression to the leading side and a negative suction-like contribution to the trailing side. To obtain a physical force, a scalar radial profile must be connected to a direction, area/volume element, and units; those conventions are not fixed by the dimensionless Node 02 proxy.

Necessary closure before physical use: define a covariant stress-energy tensor for the foam, an exchange four-current, an equation of state (including radial/tangential stresses), and a dimensional coupling from the Hawking energy flux to the density/force sector.

2. Hawking flux and radial compressive density profile

For a non-rotating, uncharged Schwarzschild black hole of mass M_{BH}, the stipulated baseline expressions are:

$$T_H = \hbar\, c^3 / (8\,\pi\, G\, M_{BH}\, k_B)$$

$$L_H = \hbar\, c^6 / (15360\,\pi\, G^2\, M_{BH}{}^2)$$

The following radial flux is the user-specified ansatz, with a core cutoff at r_{core}:

$$\Phi_H(r) = \hbar\, c^4 / (61440\,\pi^2\, G^2\, M_{BH}{}^2\, r^2) \cdot \Theta(r - r_{core})$$

Popcorn density and effective coupling

$$\rho_{popcorn}(r) = \rho_0\,(1 + r/\lambda_{coh})^{-\gamma_{vac}} \exp(-r/R_{cutoff})$$

$$\rho_{eff}(r) = \rho_{popcorn}(r) + \tau_{relax}\,\Phi_H(r)/(r_{core} + r)$$

Reference parameters: $\rho_0 = 7.02 \times 10^{-30}$ g cm^{-3}; $\lambda_{coh} = 12.0$ kpc; $\gamma_{vac} = 1.0$; $r_{core} = 4.5$ kpc. The project solver specifies $R_{cutoff} = 50$ kpc. Unless stated otherwise, r and the length scales must use the same units.

Dimensional-consistency flag (important). Isotropic luminosity dilution gives $L_H/(4\pi r^2) = \hbar c^6/(61440\pi^2 G^2 M_{BH}{}^2 r^2)$, whereas the supplied Φ_H numerator contains c^4. Thus the specified profile differs by c^2 from $L_H/(4\pi r^2)$ and does not have energy-flux dimensions under SI conventions. It is preserved here as requested, but must be reconciled before physical interpretation. Likewise, τ_{relax} must carry explicitly chosen units (and likely a c^2 conversion) so the second term has mass-density units.

The Hawking temperature and luminosity are idealized semiclassical results; actual spectral emission involves greybody factors and depends on black-hole parameters. No black-hole mass or measured Hawking signal is supplied here.

3. Front hydrodynamic decomposition & sign transition

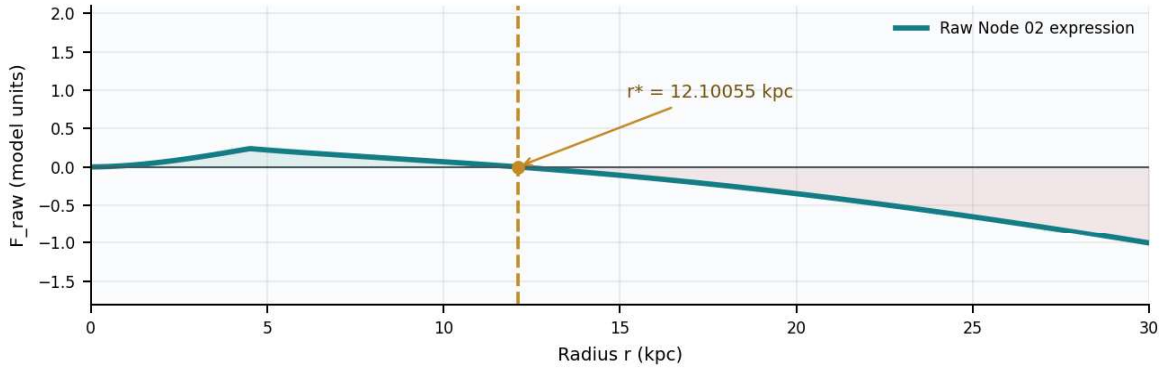
The proposed leading and trailing contributions are written as radial scalar profiles:

$$F_{front}(r) = +\beta_{front} \ln(1 + r/r_{core}) [1 - \exp(-r/\lambda_{coh})] > 0$$

$$F_{back}(r) = -\beta_{back} \max(0, r - r_{core}) < 0$$

For numerical comparison with the project source (nodes/active_tensor_foam.py), the un-clamped raw Node 02 profile is:

$$F_{friction,raw}(r) = 1.1 \ln(1 + r/4.5) [1 - \exp(-r/12.0)] - 0.12 \max(0, r - 4.5)$$



The zero is obtained by solving $F_{friction,raw}(r)=0$ numerically with a bracketed root finder: $r_{transition}$ = 12.10054930 kpc (approximately 12.10055 kpc). The root is computed directly from the stated equation. The sign is positive interior to the root (apart from r=0), and negative immediately outside it. The root marks a change in this proxy, not a demonstrated physical transition in NGC 6503.

Production distinction and control mapping

The raw Node 02 expression is explicitly sign-changing; the source code documents it as a comparison profile, not the production drag law. Node 05 instead uses a non-negative exchange coefficient Γ(r) by construction; with the implemented force convention, F = −ΓΔv, the sign is governed by relative velocity. Accordingly, this addendum must not present the outer negative raw profile as a production anti-damping current.

PID mapping (stipulated): $K_P = 0.92$, $K_I = 0.35$, $K_D = 0.18$

The PID values are a project mapping only: no controller transfer function, sampling interval, saturation, stability analysis, or fitted calibration is specified in this addendum. The coefficients should not be interpreted as universal physical constants.

4. Consistency with project standards & boundary conditions

The following conditions are treated as constraints on any future field-level realization. They are requirements, not proofs that the proposed model satisfies them.

Stress-energy covariance: $\nabla^{\mu} T_{\mu\nu}^{\text{popcorn}} = 0$
Asymptotic screening: $\lim_{r\to\infty} \rho_{\text{popcorn}}(r) = 0$
Anisotropy bound: $\lvert p_r - p_t \rvert / (\rho\, c^2) \le \Gamma_{\max}$

The specified exponential cutoff ensures the stated popcorn density tends to zero for positive finite R_{cutoff}. The anisotropy inequality cannot be evaluated until p_r, p_t, ρ, and $\Gamma_{\max}$ have operational definitions and units. Covariant conservation requires a tensor-level model, not only a radial density law. Annex A2 further requires causal horizon/microcausality control and appropriate scale decoupling.

Theory / production separation

Production behavior. Node 05 maintains a positive semi-definite exchange coefficient/current $\Gamma(r) \geq 0$ by construction. Theoretical extension. This addendum sketches a reactive, non-dissipative hydrodynamic interpretation and retains the sign-changing Node 02 raw profile for transparency and analysis. The theoretical language does not replace or silently modify production code.

Open verification items

Item	Required next step
Flux dimensions	Resolve c^4 versus c^6 normalization and define τ_{relax} units.
Field closure	Construct a covariant action/stress tensor and exchange current; test conservation and stability.
Causality / anisotropy	Specify horizon/coherence bound, stresses, and numerical $\Gamma_{\max}$.
Observation	Fit and report SPARC residual tests with uncertainties and comparison baselines (Annex A2 §1).
Controller	Document discrete/continuous PID law, units, gains, and stability domain.

Source basis & notation

Project-source cross-check: nodes/active_tensor_foam.py (Node 02 raw friction and Node 05 non-negative exchange coefficient); nodes_analysis/Annex_A2_Vacuum_Popcorn_Boundary_Conditions.md (analytical-hypothesis status and boundary constraints); Annex A — Vacuum-Popcorn Physical Constraints. Numerical root: direct solution of the raw Node 02 equation, independently evaluated from its stated coefficients. Symbols $\hbar$, c, G, k_B, M_{BH}, r, and Θ denote the reduced Planck constant, speed of light, Newton constant, Boltzmann constant, black-hole mass, radial coordinate, and Heaviside step function.

End of addendum • Document ID QVEX-ANNEX-A-HF-01 • October 2026